



Shivam Prakash

Bachelor of Technology
Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning)

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Location: Bhopal, Madhya Pradesh

Professional Summary

Analytical CS Graduate with project development experience at Bindisa Agritech. Proficient in Python, DSA, SQL, and scalable software development. Recognized for strong logical-mathematical aptitude and a disciplined approach to the SDLC, eager to contribute to company's Agile teams by developing high-impact software solutions.

Academic Qualifications

Year	Degree/Certificate	Institute	CGPA/%
2022 – 2026	B.Tech	Oriental Institute of Science and Technology, Bhopal	7.94
2022	CBSE (XII)	Creane Memorial High School, Gaya	70.00%
2020	CBSE (X)	Creane Memorial High School, Gaya	88.17%

Work Experience

- Technical Research and Development Intern — Bindisa Agritech Pvt. Ltd.** *July – Dec 2025*
 - Served as a **key innovator** by researching and proposing novel agritech solutions, including **automated scarecrow technology** and smart pest-deterrent systems.
 - Collaborated** on the development of a Soil Quality Check and Enhancement Kit by researching the integration of **IoT sensors** and information on crop and soil types.
 - Conducted technical research** on AI/ML applications in agriculture, providing data-driven insights to optimize crop suitability and moisture prediction models.
 - Analyzed and recommended** modern technological solutions to bridge the gap between complex agritech and practical field implementation for farmers.

Key Projects

- Autonomous Bio-Acoustic Deterrent System** *Oct 2025 – Dec 2025*

Tech Stack: Python, YOLOv8, OpenCV, PyAudio, MQTT, LoRaWAN

 - Leading a cross-functional team to develop an edge-computing solution for real-time crop protection, achieving species-specific animal detection with < 250ms latency.
 - Built a Python-based sound system that generates specific noises to scare away animals based on how accurately the AI identifies the species.
 - Implementing a low-bandwidth communication protocol using MQTT and LoRaWAN to ensure reliable telemetry and alert delivery from remote field nodes to a central dashboard.
- Real-time Indian Sign Language (ISL) Translator** *Aug 2025 – Oct 2025*

Tech Stack: Python, Flask, TensorFlow, OpenCV, MediaPipe

 - Engineered an end-to-end web application using **Flask** to provide real-time Indian Sign Language (ISL) translation via webcam feed and integration a **Computer Vision** pipeline for hand landmark extraction
 - Implemented advanced image preprocessing techniques using OpenCV to handle varied lighting conditions and background noise in real-time video streams.

Technical Skills

- Languages & Database:** Python, SQL, MySQL
- Core Computer Science:** DSA, DBMS, OOPS, AI, ML
- Developer Tools:** Visual Studio Code, Antigravity, Git/GitHub, Docker
- Generative AI Concepts:** Prompt Engineering, Retrieval-Augmented Generation (RAG) basics
- Soft Skills:** Adaptability, Problem Solving, Time Management, Root Cause Analysis, Decision Making

Certifications & Achievements

- Python Essentials 1 & 2** – Cisco Networking Academy (NETACAD)
- Databases and SQL** – Coursera (Guided by Meta/IBM)
- Python for Data Science** – NPTEL (Elite Certification)